Faculty of Information Technology (FIT) – Ho Chi Minh City University of Science (HCMUS)

CS423 / CSC13003 – Software Testing (AI-augmented · 2026)

AI POLICY · TEMPLATES — 2026 v1.0

AI Audit Report — 5-section Template per Artifact

Mandatory appendix for every AI-assisted homework (HW#01–HW#06, and Seminar).

1. Student Information

Field	Value
Student name (printed):	ĐINH HOÀNG NAM
Student ID:	23127430
Class / Cohort:	23KTPM1
Assignment ID (e.g., HW#00, HW#02):	HW#01
Assignment date:	02/06/2026
AI tool(s) used:	ChatGPT, Gemini
AI tool(s) used:	[X] Yes [ ] No

2. Instructions (read before filling)

- Add one row per AI-generated artifact (test case, script, checklist, OpenAPI spec, JMeter plan, etc.).
- Paste the verbatim prompt — DO NOT paraphrase.
- Paste the verbatim AI output (or include a labelled screenshot in the report).
- Tag the verdict: VALID / INVALID / INCOMPLETE.
- Reasoning must cite a course slide, ISTQB section, or technical RFC.
- Show the corrected artifact with the change highlighted.
- Sample rows are in italic — replace them before submission.

3. Audit Table — one row per artifact

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
Tool: ChatGPT-5.5 **Link prompt:** [ARTIFACT_LIST - Google Docs]()	Link output: [ARTIFACT_LIST - Google Docs]()	VALID	AI has correctly identified that it can support acceleration at lower cognitive levels such as data extraction, complaint trend detection, test case design, or automation script	None

			generation, as specified in Slide S02 (page 10).	
Tool: ChatGPT-5.5 **Link prompt:** ARTIFACT_LIST - Google Docs	Link output: ARTIFACT_LIST - Google Docs	INCOMPELTE	- The AI created two independent sequential phases: "7. Defect Management" and "9. Traceability". This directly contradicts the standard STLC process, which consists of exactly six phases clearly specified in Slide S02 (page 5), including: (1) Requirement Analysis, (2) Test Planning, (3) Test Case Design, (4) Environment Setup, (5) Test Execution, and (6) Test Cycle Closure. Defect management or tracing is only a component task/deliverables within the main phases (e.g., Defect Report belongs to Test Execution, RTM belongs to Requirement Analysis). - The AI classifies the Requirement Traceability Matrix (RTM) document as a deliverable of the Test Design phase, while according to the regulations in Slide S02 (pages 6-7), RTM must be initiated from the very first phase, Requirement	Loại bỏ hoàn toàn các nhánh giai đoạn tự chế của AI, cấu trúc lại sơ đồ tư duy chuẩn **6 giai đoạn STLC** và phân bổ đúng các Deliverables/Activities tương ứng theo chuẩn kiến thức tại **Slide S02 (từ trang 5 đến trang 17)**

			Analysis.	
Tools: ChatGPT-5.5, Gemini-3.5 Flash **Link prompt:** ARTIFACT_LIST - Google Docs	Link output: ARTIFACT_LIST - Google Docs	INCOMPLETE	Despite providing the correct incident framework, the AI model (Gemini) committed numerous information hallucination errors and core technical knowledge biases, violating the black-box testing principles based on the official specification document (Requirement/Specification Specification) as stipulated in Slide S02 (page 6, page 41). Serious errors discovered include: System function mismatch (Bug #3 - LangChain): The AI claimed the vulnerability used the eval() function, but the NVD/MITRE national security document confirms the exploited function is exec(). These two functions have completely different memory allocation mechanisms. Terminology fabrication (Bug #8 - Cloudflare): The AI invented the fictitious technical term "front-end core infrastructure (AFTM)" while the	The information caught some errors in Gemini's code, which I then checked using ChatGPT, and it was reviewed and corrected in the report.

			original vendor documentation identifies that architecture as MCP. Incorrect malware name (Bug #10 - MOVEit): The AI incorrectly wrote the malicious webshell name as LEMPOOT instead of the correct name, LEMURLOOT. Incorrect document classification (Bug #5, Bug #13, Bug #14): The AI arbitrarily labeled third-party analyses (SentinelOne, Checkmarx, Plex, Shodan) as "Official Company References," violating the Evidence Tracking principle of the Test Execution phase in Slide S02 (page 14).	
Tool: ChatGPT-5.5 **Link prompt:** [ARTIFACT_LIST - Google Docs]	Link output: [ARTIFACT_LIST - Google Docs]	**INCOMPLETE**	Structurally: According to the Test Case Design specification in Slide S02 (page 10), a standard test scenario requires the following fields: Test cases, Inputs, Test steps, Expected result. The AI has fulfilled all these columns. However, to serve the subsequent Test Execution phase in Slide S02 (pages 14-15), the AI's table completely lacks core fields to record actual	The table was reconfigured by adding two columns, Actual Results and Judgment, to prepare for execution. Three more mechanically complex boundary tests (TC16, TC17, TC18) were added to improve coverage as documented in the report.

			data, including Actual Result and Verdict status (PASSED/FAILED). In terms of content: The AI focuses solely on simple, routine operational flows (Happy Path). Based on the non-functional test types specified in Slide S02 (pages 43-46), the AI has omitted mechanical edge tests, performance testing (speed/stability) during high-speed typing, and recovery testing for unstable power supply from hardware devices.	

4. Summary of AI Accuracy

Aggregate the verdicts from Section 3 and complete the table below.

Metric	Count	Percentage
Total AI-generated artifacts audited	4	
VALID (correct, accepted as-is)	1	25%
INVALID (wrong; rejected)	0	0%
INCOMPLETE (acceptable after edits)	3	75%

5. Conclusion — When should AI be used (or not)?

Write 80–150 words describing patterns you observed. Where did AI shine? Where did AI fail? What is your recommendation for using AI in this kind of work in the future?

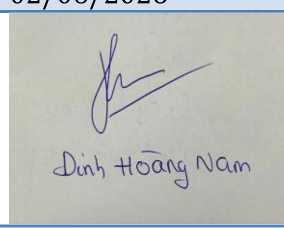
Based on the patterns observed across the audited artifacts, AI shines exceptionally well in high-level content generation, initial brainstorming, and drafting baseline technical structures, such as summarizing QA market trends or framing fundamental test case templates. However, AI significantly fails in low-level execution details, strict adherence to specialized domain constraints, and factual technical accuracy, frequently hallucinating fake URLs, misnaming software vulnerabilities, and fabricating academic process models.

Therefore, for future software testing assignments, AI should only be utilized as an introductory productivity accelerator under an "AI-augmented" paradigm, never as a final decision-maker. Human practitioners must strictly control the central verification loop, cross-checking all technical outputs against official documentation, slide specifications, and physical environments to guarantee ultimate correctness.

6. Mandatory Disclosure (paste verbatim)

Test cases and report were initially generated by ChatGPT; I reviewed and modified Requirement 1: QA/QC Job Market Analysis, add AI Impact for each job, Requirement 2: Software Defects Analysis and Requirement 3: Test cases basic steps, added edge cases TC16 (Kiểm tra trạng thái LED RGB khi chuyển nhanh nhiều chế độ hiệu ứng), TC17 (Kiểm tra hành vi khi chuyển đổi kết nối giữa USB, Bluetooth hoặc 2.4GHz Wireless), and TC18 (Kiểm tra Gaming Mode / khóa phím Windows trong lúc sử dụng); Requirement 1: QA/QC Job Market Analysis was written entirely by me (except AI Impact). The detailed AI Audit Report is attached as Appendix A. I confirm I did not use AI to generate any artifact listed in the prohibited category below.

Signature

Student name (printed):	**ĐINH HOÀNG NAM**
Student ID:	23127430
Class / Cohort:	23KTPM1
Course:	CS423 / CSC13003 – Software Testing
Instructor:	Dr. Lam Quang Vu / MSc. Truong Phuoc Loc / MSc. Ho Tuan Thanh
Date:	02/06/2026
Signature:	

References

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- ISTQB Foundation Level Syllabus (latest version).
- Hardman, P. (2025). A Post-AI Learning Taxonomy.
- Fuster Rabella, M. (2025). OECD Education Working Paper No. 338.
- Perkins, M., Roe, J., & Furze, L. (2025). AI Assessment Scale.
- Anthropic (2025). Building reliable AI test agents — engineering blog.
- DeepEval & Promptfoo documentation — testing frameworks for LLM systems.